

Chapter 2

Lines and Parabolas

This week studies the two simplest non-constant function shapes: lines and parabolas. They are familiar from algebra, but the goal here is to see the geometry that the formulas are recording.

A line records a fixed rate of change. Every step to the right raises or lowers the output by a constant amount, and geometrically that means the right triangles riding along the line are all similar. A parabola records a different kind of structure: it is the curve associated with a focus, an axis, and a symmetry that makes parallel rays converge to a single point. These two shapes become the basic models for constant change and changing change.

2.1 Lines as Families of Similar Triangles

Slope is often introduced as a formula, but the reason the formula works is geometric. Take any two points on a non-vertical line and drop the right triangle between them: run horizontally, then rise vertically to meet the line. Choose a different pair of points and build the same kind of triangle.

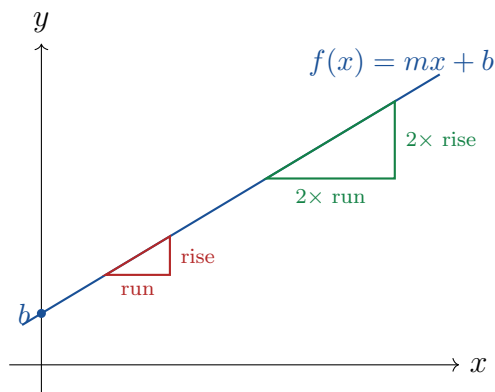
The two triangles have the same angles, because the line meets every horizontal at one fixed angle. Therefore the triangles are similar. Similar triangles have proportional sides, so the ratio of rise to run is identical no matter which pair of points generated it. That common ratio is the slope.

Definition: Slope. For two points (x_1, y_1) and (x_2, y_2) with $x_1 \neq x_2$, the *slope* of the line through them is

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\text{rise}}{\text{run}}.$$

The picture. A line is a family of similar right triangles stacked along it. Each triangle has a horizontal leg (the run) and a vertical leg (the rise), and the slope m

is the fixed ratio rise/run shared by all of them. Doubling the run doubles the rise; the shape of the triangle never changes, only its size.



Once the slope is known, the line's tilt is known, but the line can still slide up or down. The number b records where that tilted line crosses the y -axis, the output at input 0. The slope sets the tilt; the intercept sets the height. Together they fix the line completely, which is the content of the slope-intercept form.

Key idea. The slope-intercept form of a line is

$$f(x) = mx + b,$$

where m is the common rise-over-run ratio of every similar triangle along the line and $b = f(0)$ is the height at which the line crosses the y -axis.

Example 2.1.1. (*reading slope and intercept*) The line $f(x) = \frac{3}{4}x - 2$ has slope $\frac{3}{4}$: every run of 4 produces a rise of 3. It crosses the y -axis at $(0, -2)$. From the intercept, stepping right 4 and up 3 reaches $(4, 1)$, a second point on the line.

The coefficient $\frac{3}{4}$ is therefore not just a number attached to x . It is the shape of every rise-run triangle on the line.

Common pitfall. Slope is rise over run, not run over rise. The line $f(x) = \frac{3}{4}x - 2$ climbs 3 for every 4 across, a gentle rise. Inverting the ratio to $\frac{4}{3}$ describes a steeper line and a different function.

2.2 Two Points Determine a Line

A point cloud of exactly two points (with different x -coordinates) determines one line and only one. This is the linear version of fitting a model to data: there are two parameters, m and b , and two points give enough information to determine them.

The construction runs in three stages. First compute the slope from the two points. Then use one of the points to anchor the line in place. Finally rearrange the result to slope-intercept form, where the slope and intercept can be read directly. The slope comes straight from the definition. With slope and one known point (x_1, y_1) in hand, every other point (x, y) on the line must produce that same slope against (x_1, y_1) :

$$\frac{y - y_1}{x - x_1} = m.$$

Clearing the denominator gives the point-slope form. This form is useful because it keeps visible the two pieces of information we actually know: one point and one slope.

Key idea. The point-slope form of the line through (x_1, y_1) with slope m is

$$y - y_1 = m(x - x_1).$$

Solving for y converts it to slope-intercept form $y = mx + b$.

Example 2.2.1. (*from two points to $f(x) = mx + b$*) Find the line through $(2, 1)$ and $(6, 4)$.

We will separate the work into slope, point-slope form, and slope-intercept form. Keeping those stages separate makes the method easier to check.

Slope. $m = \frac{4 - 1}{6 - 2} = \frac{3}{4}.$

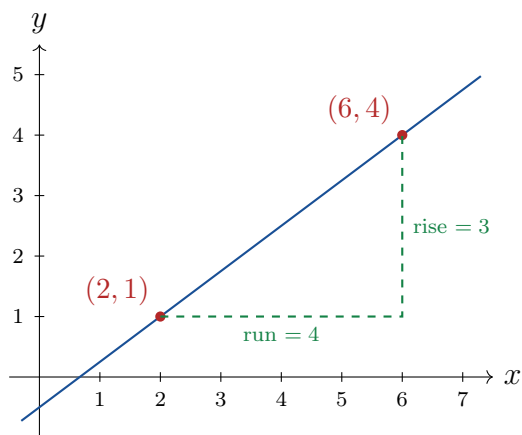
Point-slope. Using $(2, 1)$,

$$y - 1 = \frac{3}{4}(x - 2).$$

Slope-intercept. Expand and solve for y :

$$y = \frac{3}{4}x - \frac{3}{2} + 1 = \frac{3}{4}x - \frac{1}{2}, \quad \text{so} \quad f(x) = \frac{3}{4}x - \frac{1}{2}.$$

Both original points check: $f(2) = \frac{3}{4} \cdot 2 - \frac{1}{2} = \frac{3}{2} - \frac{1}{2} = 1$ and $f(6) = \frac{3}{4} \cdot 6 - \frac{1}{2} = \frac{9}{2} - \frac{1}{2} = 4$.



Common pitfall. The slope must use the coordinates in consistent order: the y -difference and the x -difference both run from the same starting point. Subtracting $y_2 - y_1$ over $x_1 - x_2$ flips the sign and tilts the line the wrong way.

Example 2.2.2. (*a horizontal and a vertical case*) Through $(1, 3)$ and $(5, 3)$ the slope is $\frac{3-3}{5-1} = 0$, giving $f(x) = 3$, a horizontal line. A zero slope means the output does not change as the input changes.

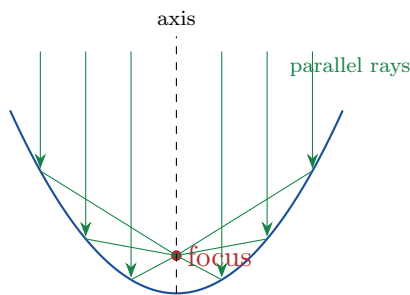
Through $(2, 1)$ and $(2, 7)$ the run is $2 - 2 = 0$, so the slope formula divides by zero. The relation $x = 2$ is a vertical line, which fails the vertical line test and so is not a function. This is not a line with an unusual slope; it is a relation that cannot be written as $y = f(x)$.

2.3 The Parabola as a Focusing Curve

A parabola is not merely a bent line. It is the next basic shape after a line, and its geometry is richer. Algebraically, it comes from a quadratic function. Geometrically, it is the curve defined by a focusing property: every ray traveling parallel to its axis reflects off the curve and passes through a single point, the focus.

This is why headlight reflectors, satellite dishes, and solar collectors are built using parabolic geometry. The same curve can concentrate parallel incoming energy at one point, or, run in reverse, send energy from the focus out as a parallel beam.

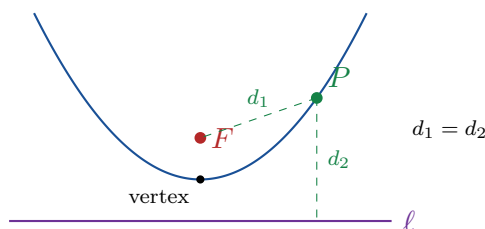
The picture. Parallel rays strike the inside of a parabola and all bounce to the focus. No other curve collects parallel rays to a single point. The algebraic parabola $y = ax^2$ and the optical reflector are the same object viewed two ways.



The focusing property has an equivalent point-and-line description. This definition looks different from the optical picture, but it describes the same curve and gives a clean way to construct it from distances.

Definition: Parabola. Fix a point F (the *focus*) and a line ℓ not through F (the *directrix*). The *parabola* is the set of all points equidistant from F and ℓ .

So a point and a line determine a parabola through the equidistance condition, just as two points determine a line. A point P lies on the parabola exactly when its distance to the focus equals its perpendicular distance to the directrix. The vertex sits halfway between focus and directrix, and the axis is the perpendicular line from the focus to the directrix.



2.4 Three Points Determine a Parabola

A vertical-axis parabola is the graph of a quadratic function

$$f(x) = ax^2 + bx + c, \quad a \neq 0.$$

The three numbers a , b , and c control the parabola, so three pieces of independent information are needed to determine them.

Three points with distinct x -coordinates give three equations. Solving that system fixes the three parameters uniquely. In this sense, three points determine a vertical-axis parabola in the same way that two points determine a line.

Key idea. Two points (distinct x) determine a unique line, $f(x) = mx + b$, with two parameters. Three points (distinct x) determine a unique vertical-axis parabola, $f(x) = ax^2 + bx + c$, with three parameters. The count of points matches the count of parameters.

Example 2.4.1. (*fitting a parabola to three points*) Find the quadratic through $(0, 1)$, $(1, 0)$, and $(2, 3)$.

The unknowns are a , b , and c . Each point gives one equation by substituting its x - and y -coordinates into $f(x) = ax^2 + bx + c$:

$$\begin{aligned} (0, 1) : \quad & c = 1, \\ (1, 0) : \quad & a + b + c = 0, \\ (2, 3) : \quad & 4a + 2b + c = 3. \end{aligned}$$

From the first, $c = 1$. The second gives $a + b = -1$. The third gives $4a + 2b = 2$, i.e. $2a + b = 1$. Subtracting, $a = 2$, then $b = -3$. The quadratic is

$$f(x) = 2x^2 - 3x + 1.$$

Check $(2, 3)$: $2 \cdot 4 - 6 + 1 = 3$.

This example is also a reminder that fitting a quadratic is a linear algebra problem in disguise. The equations are linear in the unknown parameters a , b , and c , even though the resulting function is quadratic in x .

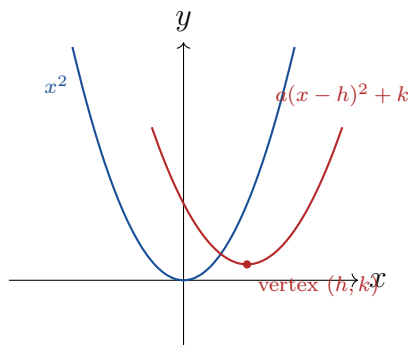
Common pitfall. The three points must have distinct x -coordinates. Two points sharing an x -value would force two outputs at one input, violating the function condition, and the system would have no solution.

2.5 Three Forms of a Quadratic

A quadratic function can be written three common ways. The forms describe the same parabola, but each one makes different information easy to see. Part of learning quadratics is learning which form to use for the question being asked.

Form	Expression	What it shows at a glance
Standard	$f(x) = ax^2 + bx + c$	y -intercept c ; leading behavior a
Factored	$f(x) = a(x - r_1)(x - r_2)$	the roots (x-intercepts) r_1, r_2
Vertex	$f(x) = a(x - h)^2 + k$	the vertex (h, k)

In all three forms the same a controls the opening: upward for $a > 0$, downward for $a < 0$, and steeper as $|a|$ grows. The vertex form is the transformation form from Week 1 with parent $f(x) = x^2$: h shifts horizontally, k shifts vertically, and a scales and possibly reflects. The same parabola is being viewed through three different algebraic windows.



2.5.1 Standard to vertex: completing the square

Completing the square rewrites $ax^2 + bx + c$ as $a(x - h)^2 + k$ by reconstructing the perfect square hidden in the first two terms. The goal is not to change the function, but to reveal its vertex.

The procedure is systematic: factor a from the x -terms, add and subtract the square that completes the bracket, then collect the constant terms. **Example 2.5.1.** (*completing the square*) Convert $f(x) = 2x^2 - 12x + 19$ to vertex form.

Factor 2 from the variable terms:

$$f(x) = 2(x^2 - 6x) + 19.$$

The square completing $x^2 - 6x$ uses half of -6 , squared, which is 9:

$$f(x) = 2(x^2 - 6x + 9 - 9) + 19 = 2((x - 3)^2 - 9) + 19.$$

Distribute and collect:

$$f(x) = 2(x - 3)^2 - 18 + 19 = 2(x - 3)^2 + 1.$$

The vertex is $(3, 1)$ and the parabola opens upward, stretched by 2.

The completed square has moved the important geometry to the surface. In $2(x - 3)^2 + 1$, the numbers 3 and 1 are no longer hidden inside the expanded quadratic; they are the coordinates of the vertex.

The vertex coordinates can also be read off a formula that completing the square produces in general:

$$h = -\frac{b}{2a}, \quad k = f(h).$$

Example 2.5.2. (*vertex by formula*) For $f(x) = 2x^2 - 12x + 19$, $h = -\frac{-12}{2 \cdot 2} = 3$ and $k = f(3) = 18 - 36 + 19 = 1$. The vertex is $(3, 1)$, agreeing with the completed-square result.

2.5.2 Standard to factored: the roots

The factored form exposes the x -intercepts, the inputs where $f(x) = 0$. These are also called roots or zeros. They come from factoring directly when possible, or from the quadratic formula in general:

$$r_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

The quantity under the root, $b^2 - 4ac$, is the *discriminant*. It records how many real roots exist before we even compute them: two when positive, one (a doubled root at the vertex) when zero, and none when negative. Geometrically, it tells us whether the

parabola crosses the x -axis twice, touches it once, or misses it. **Example 2.5.3.** (*standard to factored*) Factor $f(x) = x^2 - x - 6$. Two numbers multiplying to -6 and adding to -1 are -3 and 2 , so

$$f(x) = (x - 3)(x + 2),$$

with roots $r_1 = 3$ and $r_2 = -2$. The discriminant is $(-1)^2 - 4(1)(-6) = 25 > 0$, confirming two real roots.

Example 2.5.4. (*roots by the quadratic formula*) For $f(x) = 2x^2 - 3x + 1$, the discriminant is $9 - 8 = 1$, so

$$r = \frac{3 \pm 1}{4} = 1 \text{ or } \frac{1}{2}.$$

The factored form is $f(x) = 2(x - 1)(x - \frac{1}{2})$. Expanding recovers $2x^2 - 3x + 1$.

2.5.3 Relations among the parameters

The three forms are not separate facts to memorize. They share structure, and a few relations connect their parameters directly. These relations are useful because they let us move from one piece of information to another without always expanding or factoring from scratch.

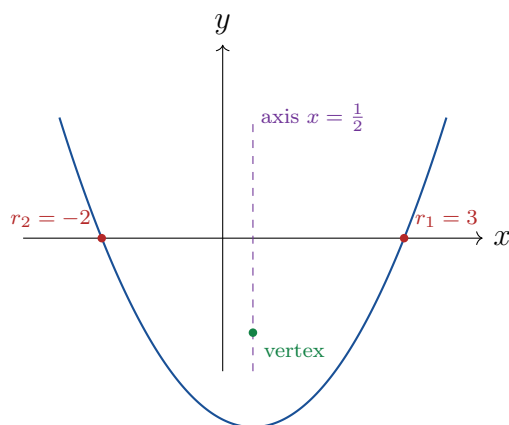
Key idea. For $f(x) = ax^2 + bx + c = a(x - r_1)(x - r_2) = a(x - h)^2 + k$:

- the vertex sits at $h = -\frac{b}{2a}$, midway between the roots: $h = \frac{r_1 + r_2}{2}$;
- the product of the roots is $r_1 r_2 = \frac{c}{a}$ and their sum is $r_1 + r_2 = -\frac{b}{a}$;
- the y -intercept is $c = f(0) = a r_1 r_2 = a h^2 + k$.

Example 2.5.5. (*reading the vertex from the roots*) The parabola $f(x) = (x - 3)(x + 2)$ has roots 3 and -2 . Their midpoint is $h = \frac{3 + (-2)}{2} = \frac{1}{2}$, so the axis of symmetry is $x = \frac{1}{2}$ and the vertex lies there. Then $k = f(\frac{1}{2}) = (\frac{1}{2} - 3)(\frac{1}{2} + 2) = (-\frac{5}{2})(\frac{5}{2}) = -\frac{25}{4}$. The vertex is $(\frac{1}{2}, -\frac{25}{4})$.

The averaging step works because the two roots lie symmetrically on either side of the axis. Once the axis is known, evaluating the function there gives the height of the vertex.

The picture. A parabola is symmetric about the vertical line through its vertex. The two roots sit at equal distances on either side of that axis, so their midpoint is the vertex's x -coordinate. Reading the vertex from the roots is just averaging.



2.6 Exercises

Exercises 2.6

- 2.6.1. Explain why the slope of a line is the same regardless of which two points are used to compute it, referring to similar triangles in your answer.
- 2.6.2. Find the equation of the line through $(-1, 2)$ and $(3, -6)$. Carry it through slope, point-slope, and slope-intercept form, and verify with both points.
- 2.6.3. A line passes through $(4, 1)$ with slope $-\frac{2}{3}$. Write its point-slope and slope-intercept forms, and find its x - and y -intercepts.
- 2.6.4. Decide whether each relation is a function and, if so, give its slope: (a) the line through $(2, 5)$ and $(2, -1)$; (b) the line through $(0, 4)$ and $(7, 4)$; (c) the line through $(1, 1)$ and $(3, 5)$.
- 2.6.5. State the focus-directrix definition of a parabola and explain, in terms of equidistance, why the vertex lies halfway between focus and directrix.
- 2.6.6. Describe the focusing property of a parabola and name two devices that exploit it. Explain which direction the rays travel in each device.
- 2.6.7. Find the quadratic $f(x) = ax^2 + bx + c$ through $(-1, 6)$, $(0, 1)$, and $(2, 3)$ by setting up and solving the linear system.
- 2.6.8. Convert $f(x) = 3x^2 + 12x + 7$ to vertex form by completing the square, and confirm the vertex with the formula $h = -\frac{b}{2a}$.

- 2.6.9. For $f(x) = x^2 - 2x - 15$, find the roots by factoring, the vertex by averaging the roots, and the y -intercept. Sketch the parabola with all three labeled.
- 2.6.10. Use the discriminant to determine the number of real roots of each: (a) $x^2 - 4x + 4$; (b) $2x^2 + x + 3$; (c) $x^2 - 5x + 2$. For any with real roots, give them.
- 2.6.11. A quadratic has roots -4 and 2 and passes through $(0, -16)$. Determine a , then write the function in all three forms.
- 2.6.12. Show, using $h = -\frac{b}{2a}$ and the root-sum relation $r_1 + r_2 = -\frac{b}{a}$, that the vertex's x -coordinate equals the average of the two roots. Explain the connection to the axis of symmetry.